

Module 5

Two-Port Networks and Network Graphs

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Cut-set matrix

A cut-set is a minimum set of branches in a connected graph that, when removed, separates the graph into two sub-graphs. The fundamental cut-set represents these sets, with each row corresponding to a fundamental cut-set and each column to a branch.

To construct the matrix, you identify a tree and its links, then create rows for the fundamental cut-sets (one per twig) and columns for all branches, using +1 or -1 to denote the branch's presence and orientation in the cut-set and 0 otherwise.

Cut-set matrix

Procedure to obtain the cut-set matrix:

Choose a tree:

Select a tree from the graph. The branches in the tree are called twigs, and the remaining branches are called links.

Identify fundamental cut-sets: For each twig, find its fundamental cut-set. This set consists of the twig and all the links that form a path between the endpoints of the twig, effectively separating the graph when the twig is removed along with the links.

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Populate the matrix: Assign a direction to each branch in the cut-set. The direction is typically determined by the twig's direction.

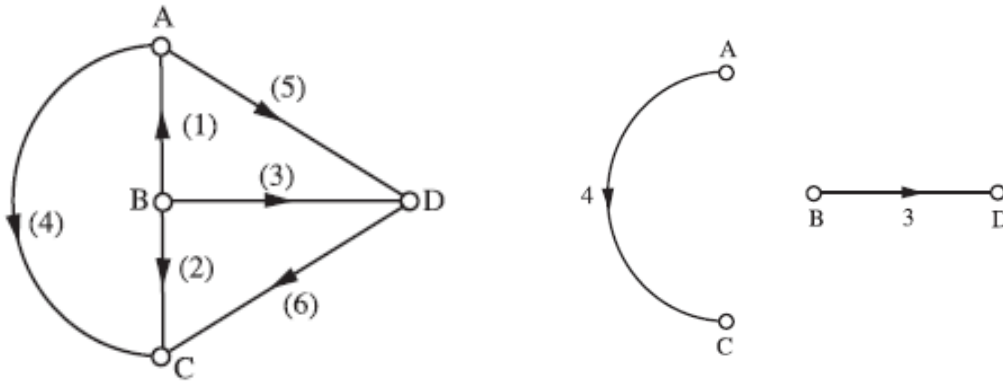
+1: If a branch is in the cut-set and its orientation is the same as the cut-set's orientation.

-1: If a branch is in the cut-set but its orientation is opposite to the cut-set's orientation.

0: If a branch is not part of that specific cut-set.

Cut-set matrix

The cut set is a minimal set of branches of the graph, removal of which cuts the graph into two parts. It separates the nodes of the graph into two groups.



The cut set consists of only one tree branch and remainders are links.

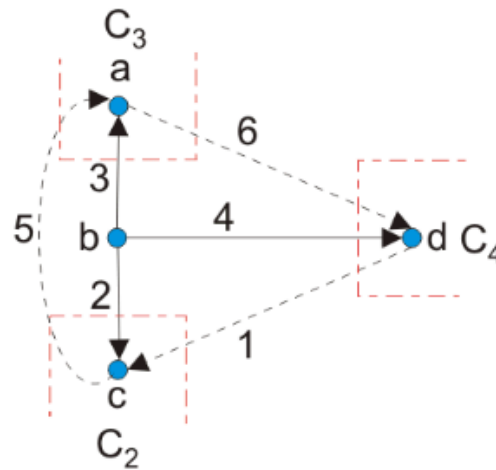
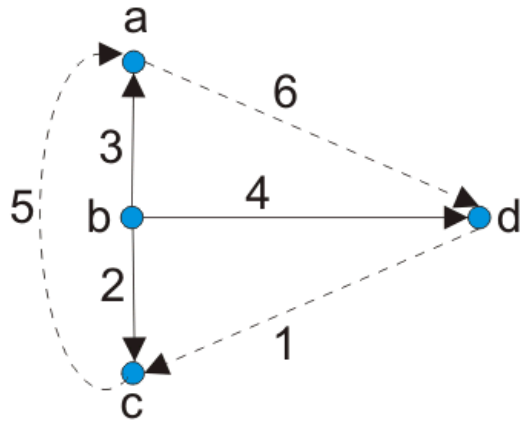
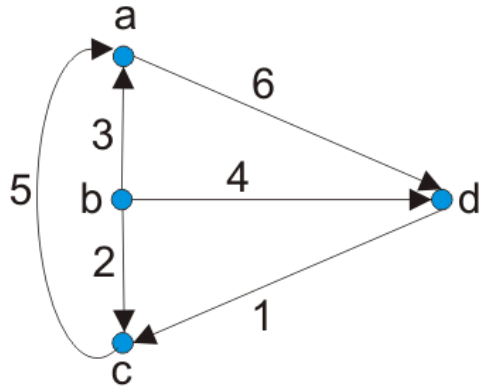
Each branch of the cut-set has one of its terminal incident at a node in one group and its other end at a node in the other group.

The orientation of the cut-set is same as orientation of tree branch.

The number of cut-sets is equal to the number of tree branches $(N-1)$.

Cut-set matrix

1. Draw the cut-set matrix for the following graph considering the tree 2, 3, 4.



Cut-set matrix

	branches					
	1	2	3	4	5	6
Cut-set 1-e2	1	1	0	0	-1	0
Cut-set 2-e3	0	0	1	0	1	-1
Cut-set 3-e4	-1	0	0	1	0	1

The columns of the cut-set schedule give branch voltage equations:

$$\begin{array}{l}
 V_1 = e_2 - e_4 \\
 V_2 = e_2 \\
 V_3 = e_3 \\
 V_4 = e_4 \\
 V_5 = e_3 - e_2 \\
 V_6 = e_4 - e_3
 \end{array}
 \left. \vphantom{\begin{array}{l} V_1 = e_2 - e_4 \\ V_2 = e_2 \\ V_3 = e_3 \\ V_4 = e_4 \\ V_5 = e_3 - e_2 \\ V_6 = e_4 - e_3 \end{array}} \right\} \text{Eqn. 1}$$

The rows of the cut-set schedule KCL equations:

$$\begin{array}{l}
 j_1 + j_2 - j_5 = 0 \\
 j_3 + j_5 - j_6 = 0 \\
 -j_1 + j_4 + j_6 = 0
 \end{array}
 \left. \vphantom{\begin{array}{l} j_1 + j_2 - j_5 = 0 \\ j_3 + j_5 - j_6 = 0 \\ -j_1 + j_4 + j_6 = 0 \end{array}} \right\} \text{Eqn. 2}$$

Cut-set matrix

The cut-set matrix can be written as the cut-set matrix (C).

$$[C] = \begin{pmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

From set of eqn. 1

$$[V_B] = [C]^T [e_N]$$

$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} e_2 \\ e_3 \\ e_4 \end{pmatrix}$$

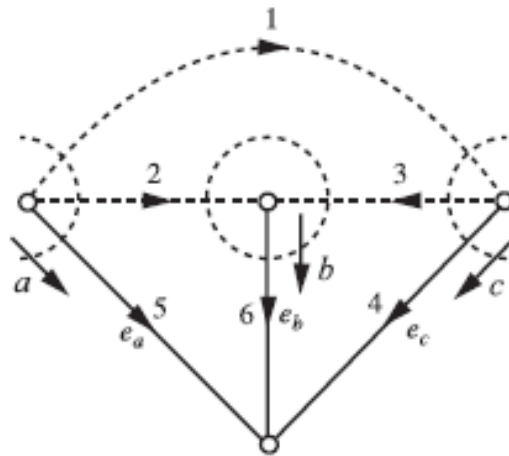
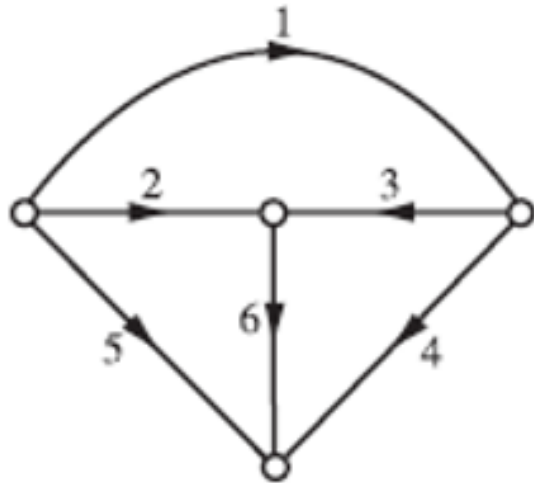
From set of eqn. 2

$$[C] [I_B] = 0$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} J_1 \\ J_2 \\ J_3 \\ J_4 \\ J_5 \\ J_6 \end{pmatrix} = 0$$

Cut-set matrix

2. Draw the cut-set matrix for the following graph with respect to the branch 4, 5, 6.



$$[C] = \begin{matrix} e_a \\ e_b \\ e_c \end{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$J_1 + J_2 + J_5 = 0$$

$$-J_2 - J_3 + J_6 = 0$$

$$-J_1 + J_3 + J_4 = 0$$

$$v_1 = e_a - e_c; \quad v_2 = e_a - e_b; \quad v_3 = e_b - e_c; \quad v_4 = e_c; \quad v_5 = e_a; \quad v_6 = e_b$$

End